

challenges/scripts/ meta_test_constitution_inheritance.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** Active paired §1.1 anti-bluff meta-test. Proves the constitution inheritance gate genuinely catches the regressions it claims to catch.

Companion (§11.4.18) to the in-source documentation block at the top of the script.

Overview / Purpose

Helix Constitution §1.1 requires every gate to have a paired mutation showing it is NOT a bluff gate. This meta-test proves `tests/constitution_inheritance_gate.sh` is genuine by **mutation testing**: for each invariant the gate asserts, it snapshots the target file, strips the anchor the gate looks for, runs the gate, and **REQUIRES** the gate to FAIL — then restores the file. A gate that still **PASSes** with its anchor removed would be a bluff (and itself a Constitution violation).

It also drives the constitution-shipped generic mutator `constitution/meta_test_inheritance.sh` against this project's gate, and finishes with a §11.4.84 working-tree quiescence check (the constitution submodule must be exactly as found — no mutation residue).

The six mutations exercised:

- strip the §11.4 section heading from `constitution/Constitution.md`;
- strip the anti-bluff covenant heading from `constitution/CLAUDE.md`;
- strip the anti-bluff covenant heading from `constitution/AGENTS.md`;
- strip the `INHERITED FROM constitution/CLAUDE.md` pointer from `CLAUDE.md`;
- strip the `constitution/AGENTS.md` reference from `AGENTS.md`;
- strip the `constitution/Constitution.md` reference from `CONSTITUTION.md`.

Plus a baseline check (gate PASSes on the un-mutated tree) and the shipped generic mutator run.

Usage

`bash challenges/scripts/meta_test_constitution_inheritance.sh`

No arguments.

Inputs

- None on the command line.
- Resolves the project root and constitution submodule from its own location; reads (and TEMPORARILY mutates + restores) constitution/Constitution.md, constitution/CLAUDE.md, constitution/AGENTS.md, and the project-root CLAUDE.md / AGENTS.md / CONSTITUTION.md.

Outputs

- Colorized (TTY) ✓ PASS / ✗ FAIL lines per mutation, a baseline line, the shipped-mutator result, a quiescence line, and a === summary: N pass, M fail ===; failures are also echoed to stderr.
- Exit code: 0 = the gate caught every mutation (gate is genuine); 1 = a mutation slipped past (bluff gate) OR a precondition/quiescence check failed.

Side-effects

- **TEMPORARILY mutates tracked governance files** — each mutation is restored immediately after the gate runs, and again via an EXIT/INT/TERM trap (crash-safe `restore_pending`), so an interrupted run never leaves residue.
- Creates and removes per-mutation `mktemp` backups.
- Ends by asserting the constitution submodule working tree is clean (§11.4.84); a dirty tree FAILs the meta-test rather than being papered over.

Dependencies

`bash`, `grep`, `git`, `mktemp`, `cp`, `rm`. Requires `tests/constitution_inheritance_gate.sh` (the gate under test) and `constitution/meta_test_inheritance.sh` (the generic mutator); a missing shipped mutator FAILs the meta-test.

Edge cases

- **Anchor absent BEFORE mutation** → the mutation is a precondition FAIL (anchor not present BEFORE mutation), never a silent skip.
- **Target file missing** → FAIL for that mutation.
- **Baseline gate FAILs on the clean tree** → reported first, so a broken gate is not mistaken for genuine mutation-catching.
- **Interrupt mid-mutation** → the trap restores the in-flight file from its backup before exit.

Related scripts

- tests/constitution_inheritance_gate.sh — the gate under test.
- constitution/meta_test_inheritance.sh — the constitution-shipped generic §11.4 mutator this test also drives.
- tests/test_constitution_inheritance.sh — the sibling inheritance test.
- Constitution §1.1 (paired-mutation discipline), §11.4.84 (working-tree quiescence), §11.4.35 (canonical-root inheritance).

Last verified

2026-07-01 — documented against the script source; sh -n / bash -n parse-clean. Executed as part of the §1.1 anti-bluff gate suite; the tree is left quiescent on exit.